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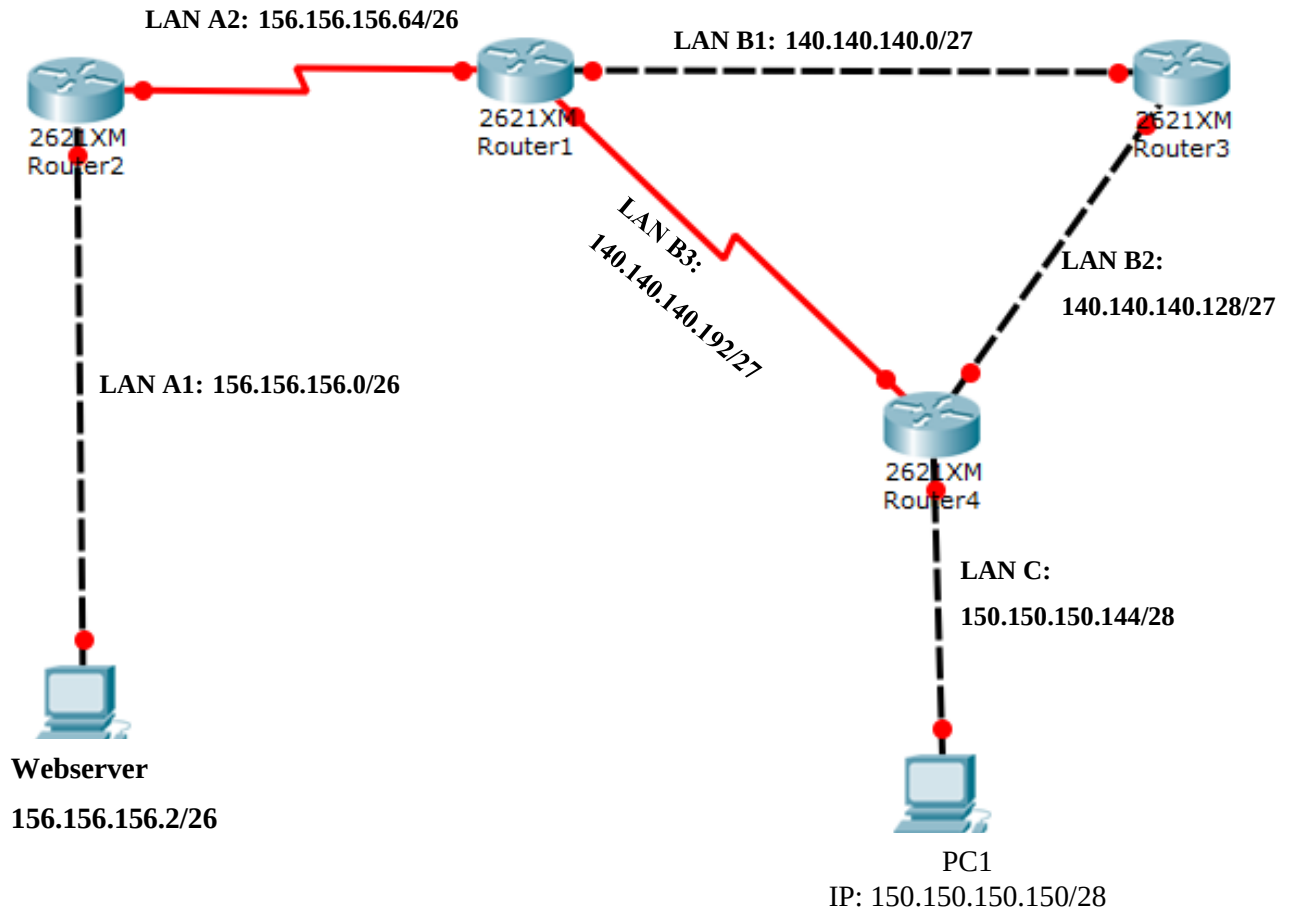
Group: M02

Declaration of own work

I, Nguyen Minh Nhut B2205896, certify that this assignment is my own work, is not copied from any other person's work.

Construct a network system as follows:

- LAN A has a single network address of 156.156.156.0/24, using static routing. LAN A is divided into 2 subnets, consisting of A1 and A2. In addition, there is a Web server running a simple webpage showing “YEAH! My name is *YOUR_FULL_NAME*” (replace *YOUR_FULL_NAME* by your full name) in LAN A1.
- LAN B1 has a network address of 140.140.140.0/27, using the RIPv2 protocol.
- LAN B2 has a network address of 140.140.140.128/27, using the RIPv2 protocol.
- LAN B3 has a network address of 140.140.140.192/27, using the RIPv2 protocol.
- LAN C includes PC1 and Router 4. The IP address of PC1 is 150.150.150.150/28.



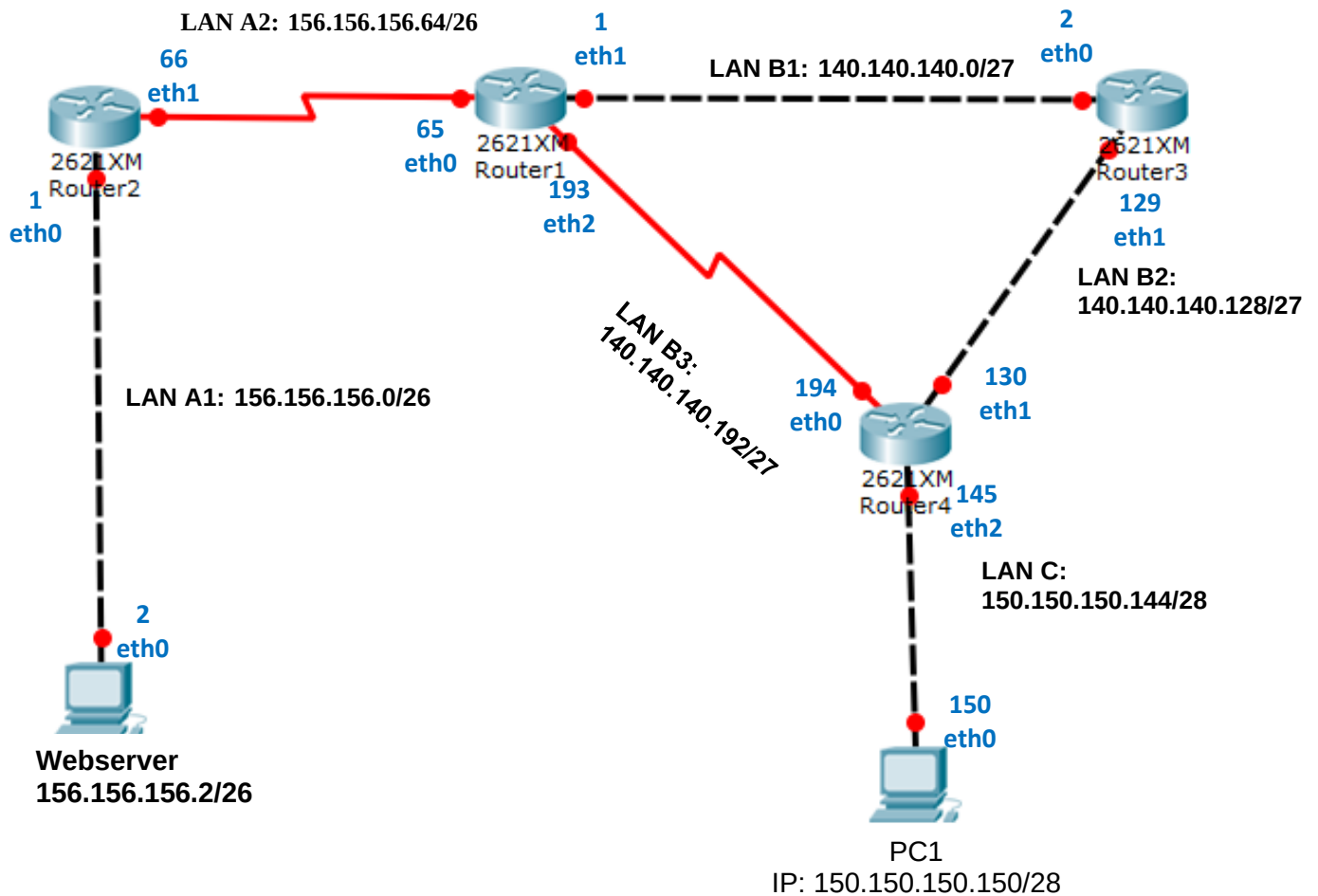
Please take screenshots showing:

1. (0,5 point) select and assign the IP addresses for all of the Ethernet interfaces.
2. (1,0 point) the directory tree structure of this network system (using the ***tree*** command).
3. (1,0 point) the content of the file *lab.conf*?
4. (5,0 points) the content of all files *. *startup*
5. (1,0 point) the contents of all files and commands you use in order to set up the web service on the web server
6. (0,5 point) the command line to check the hops for transmitting data from PC1 to the web server? List all hops between PC1 and the Web server.
7. (1,0 points) check the network system constructed (using the ***ping*** command).

*****GOOD LUCK*****

SOLUTION:

1. Select and assign the IP addresses for all of the Ethernet interfaces.



2. The directory tree structure of this network system (using the *tree* command).

```
nmn@Ubuntub2205896 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Nov 12 20:41
nmn@Ubuntub2205896: ~/CT106H/finalLab
nmn@Ubuntub2205896:~/CT106H$ mkdir finalLab; cd finalLab
nmn@Ubuntub2205896:~/CT106H/finalLab$ mkdir r2 pc1 -p r{1,3,4}/etc/quagga -p server/var/www/html shared
nmn@Ubuntub2205896:~/CT106H/finalLab$
```

```
nmn@Ubuntu2205896 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal
nmn@Ubuntu2205896: ~/CT106H/finalLab$ tree
.
├── lab.conf
├── pc1
├── pc1.startup
├── r1
│   └── etc
│       └── quagga
│           ├── daemons
│           ├── ripd.conf
│           └── zebra.conf
├── r1.startup
├── r2
├── r2.startup
├── r3
│   └── etc
│       └── quagga
│           ├── daemons
│           ├── ripd.conf
│           └── zebra.conf
├── r3.startup
├── r4
│   └── etc
│       └── quagga
│           ├── daemons
│           ├── ripd.conf
│           └── zebra.conf
├── r4.startup
├── server
│   ├── var
│   │   └── www
│   │       └── html
│   │           └── index.html
├── server.startup
└── shared

16 directories, 17 files
nmn@Ubuntu2205896: ~/CT106H/finalLab$
```

3. The content of the file lab.conf?

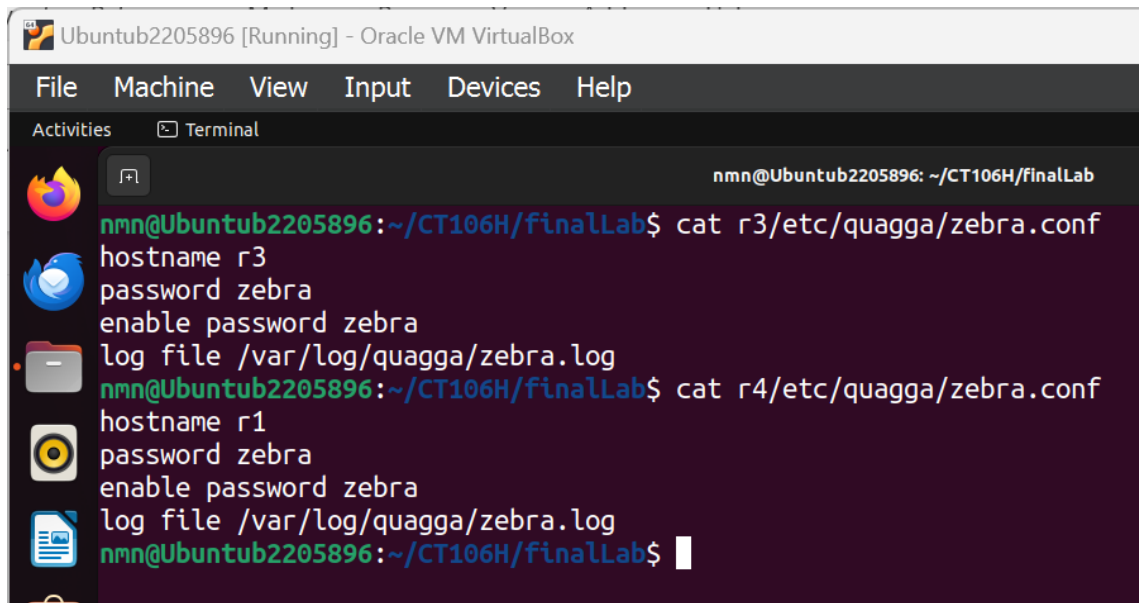
```
nmn@Ubuntu2205896 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal
nmn@Ubuntu2205896: ~/CT106H/finalLab$ cat lab.conf
server[0]=A1
r2[0]=A1
r2[1]=A2
r1[0]=A2
r1[1]=B1
r1[2]=B3
r3[0]=B1
r3[1]=B2
r4[0]=B3
r4[1]=B2
r4[2]=C
pc1[0]=C
```

4. The content of all files *. Startup

```
nmn@Ubuntub2205896: ~/CT106H/finalLab$ cat server.startup
ifconfig eth0 156.156.156.2/26 up
route add default gw 156.156.156.1
/etc/init.d/apache2 start
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat pc1.startup
ifconfig eth0 150.150.150.150/28 up
route add -net 0.0.0.0/0 gw 150.150.150.145
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r1.startup
ifconfig eth0 156.156.156.65/26 up
ifconfig eth1 140.140.140.1/27 up
ifconfig eth2 140.140.140.193/27 up
route add -net 156.156.156.0/26 gw 156.156.156.66
route add -net 150.150.150.144/28 gw 140.140.140.194
/etc/init.d/quagga start
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r2.startup
ifconfig eth0 156.156.156.1/26 up
ifconfig eth1 156.156.156.66/26 up
route add default gw 156.156.156.65
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r3.startup
ifconfig eth0 140.140.140.2/27 up
ifconfig eth1 140.140.140.129/27 up
route add default gw 140.140.140.1
/etc/init.d/quagga start
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r4.startup
ifconfig eth0 140.140.140.194/27 up
ifconfig eth1 140.140.140.130/27 up
ifconfig eth2 150.150.150.145/28 up
route add default gw 140.140.140.193
/etc/init.d/quagga start
nmn@Ubuntub2205896:~/CT106H/finalLab$
```

```
nmn@Ubuntub2205896: ~/CT106H/finalLab$ cat r1/etc/quagga/daemons
zebra=yes
bgpd=no
ospfd=no
ospf6d=no
ripd=yes
ripngd=no
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r1/etc/quagga/ripd.conf
hostname ripd
password zebra
enable password zebra
router rip
redistribute connected
network 140.140.140.0.24
log file /var/log/quagga/ripd.log
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r1/etc/quagga/zebra.conf
hostname r1
password zebra
enable password zebra
log file /var/log/quagga/zebra.log
nmn@Ubuntub2205896:~/CT106H/finalLab$
```

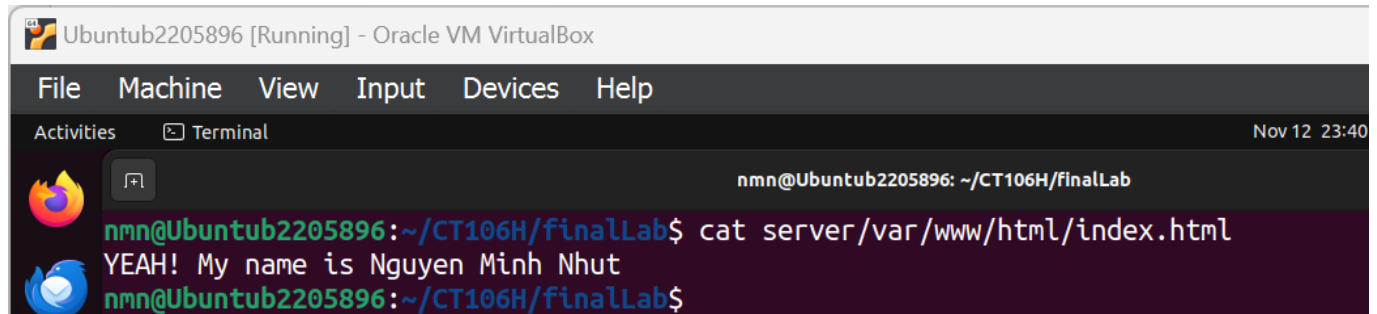
The daemons and ripd.conf of r3 and r4 are the same with r1:



```
nmn@Ubuntub2205896: ~/CT106H/finalLab
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r3/etc/quagga/zebra.conf
hostname r3
password zebra
enable password zebra
log file /var/log/quagga/zebra.log
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat r4/etc/quagga/zebra.conf
hostname r1
password zebra
enable password zebra
log file /var/log/quagga/zebra.log
nmn@Ubuntub2205896:~/CT106H/finalLab$
```

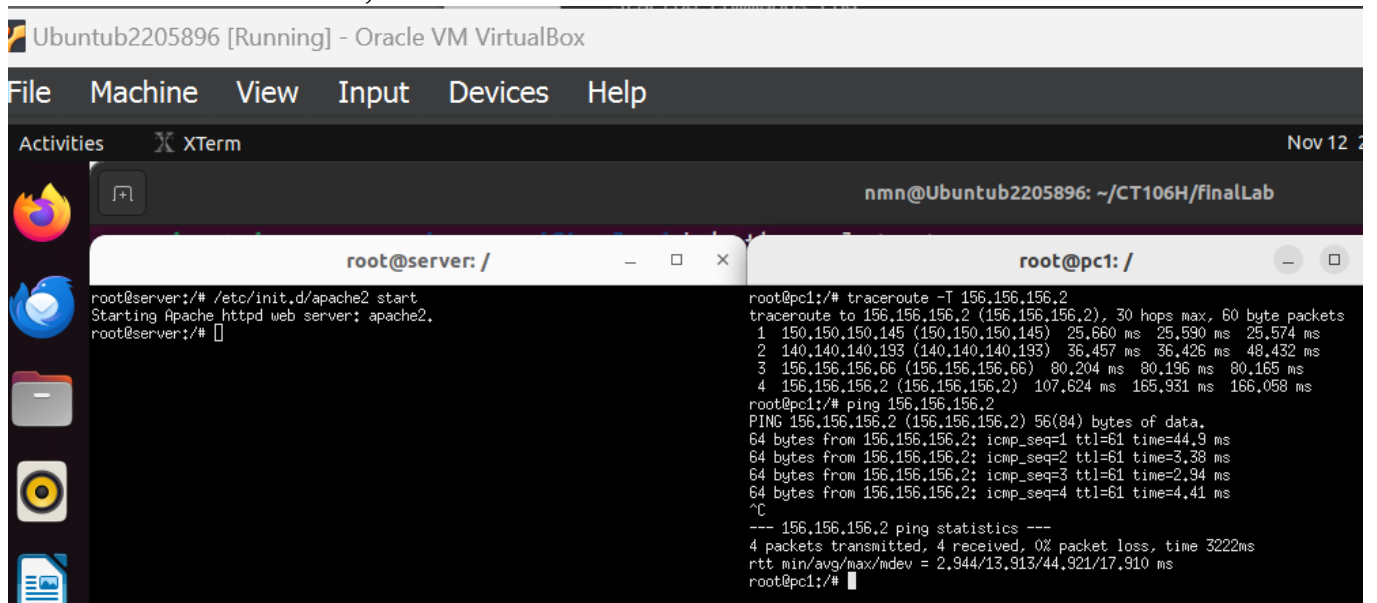
5. The contents of all files and commands you use in order to set up the web service on the web server

Content on index.html



```
nmn@Ubuntub2205896: ~/CT106H/finalLab
nmn@Ubuntub2205896:~/CT106H/finalLab$ cat server/var/www/html/index.html
YEAH! My name is Nguyen Minh Nhut
nmn@Ubuntub2205896:~/CT106H/finalLab$
```

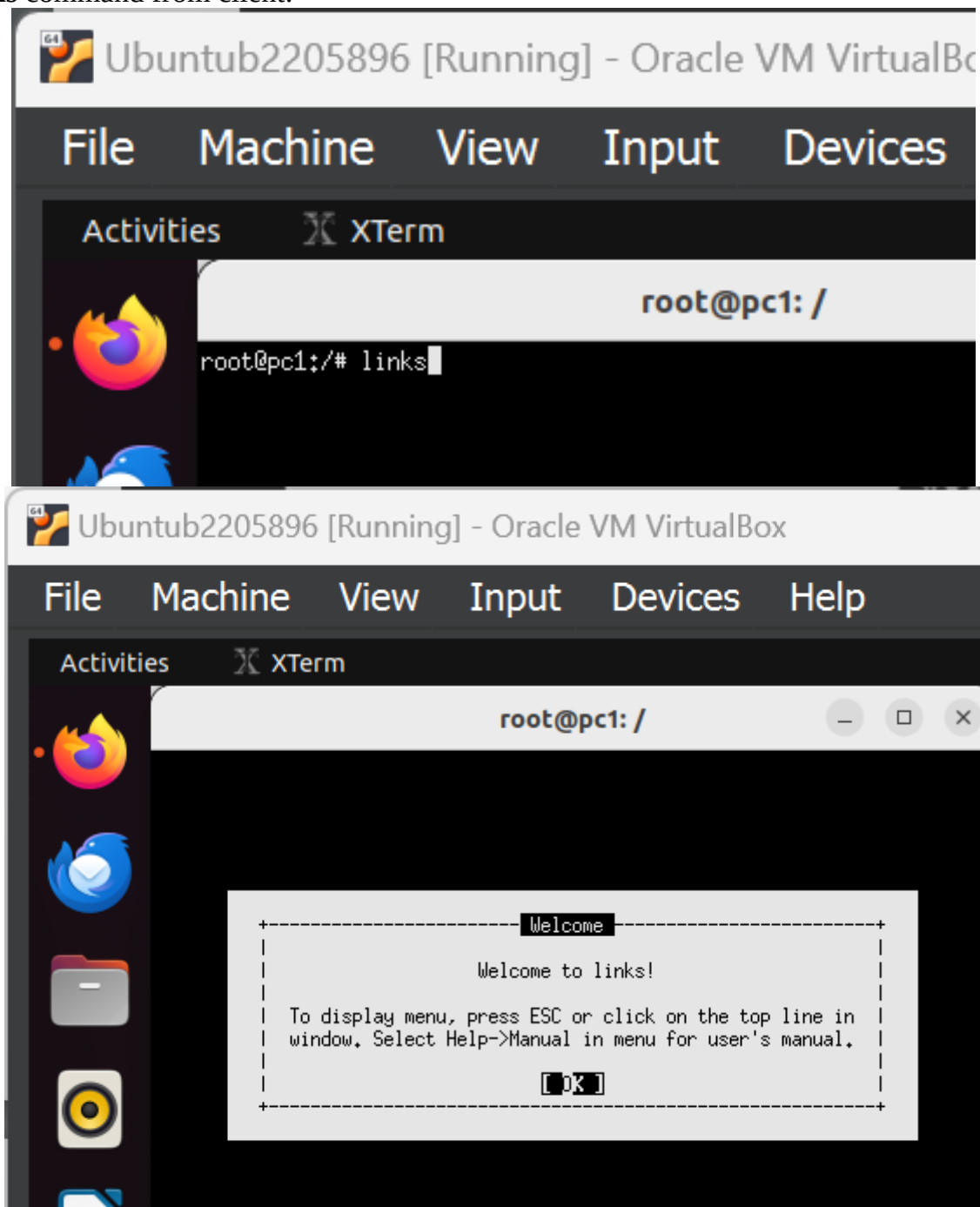
On server and from client, test connection:



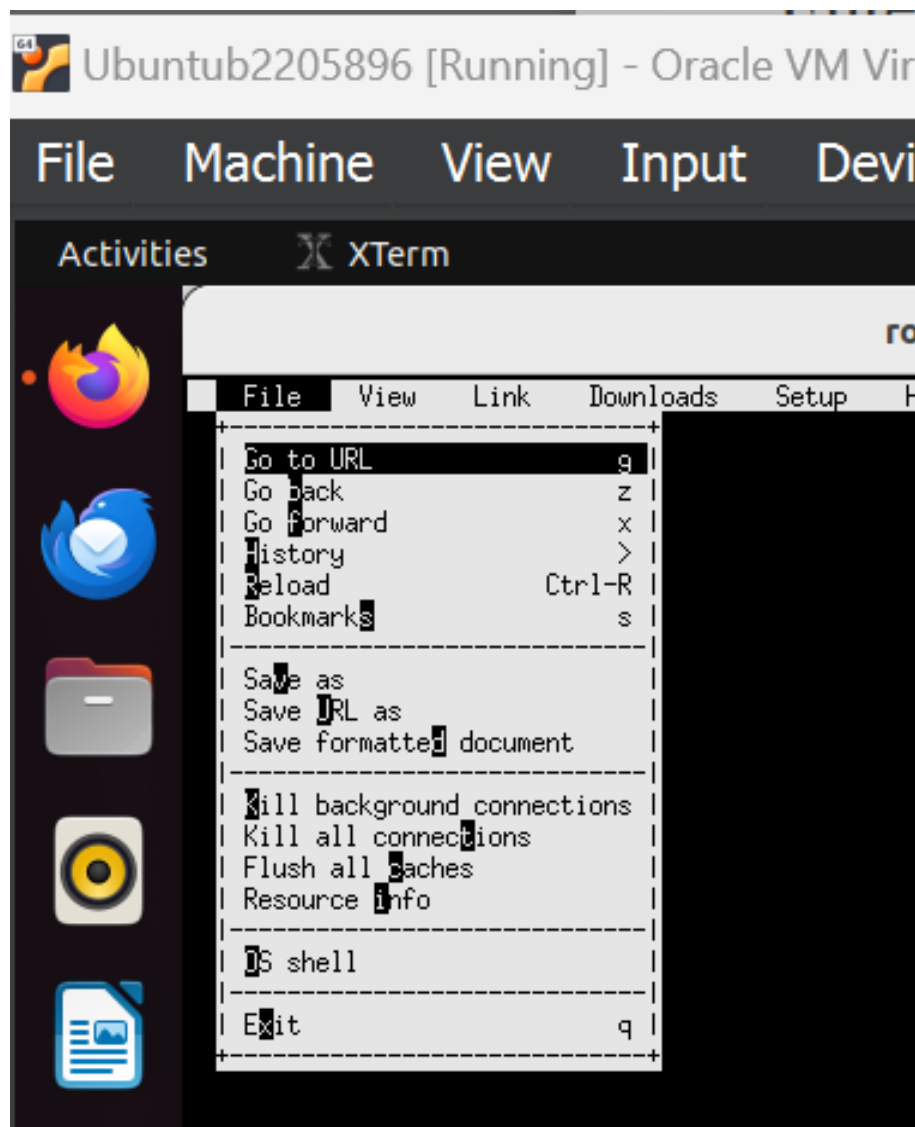
```
root@server: /
root@server:~# /etc/init.d/apache2 start
Starting Apache httpd web server: apache2.
root@server:~#

root@pc1: /
root@pc1:~# traceroute -T 156.156.156.2
traceroute to 156.156.156.2 (156.156.156.2), 30 hops max, 60 byte packets
 1 150.150.150.145 (150.150.150.145) 25.660 ms 25.590 ms 25.574 ms
 2 140.140.140.193 (140.140.140.193) 36.457 ms 36.426 ms 48.432 ms
 3 156.156.156.66 (156.156.156.66) 80.204 ms 80.196 ms 80.165 ms
 4 156.156.156.2 (156.156.156.2) 107.624 ms 165.931 ms 166.058 ms
root@pc1:~# ping 156.156.156.2
PING 156.156.156.2 (156.156.156.2) 56(84) bytes of data.
64 bytes from 156.156.156.2: icmp_seq=1 ttl=61 time=44.9 ms
64 bytes from 156.156.156.2: icmp_seq=2 ttl=61 time=3.38 ms
64 bytes from 156.156.156.2: icmp_seq=3 ttl=61 time=2.94 ms
64 bytes from 156.156.156.2: icmp_seq=4 ttl=61 time=4.41 ms
^C
--- 156.156.156.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3222ms
rtt min/avg/max/mdev = 2.944/13.913/44.921/17.910 ms
root@pc1:~#
```

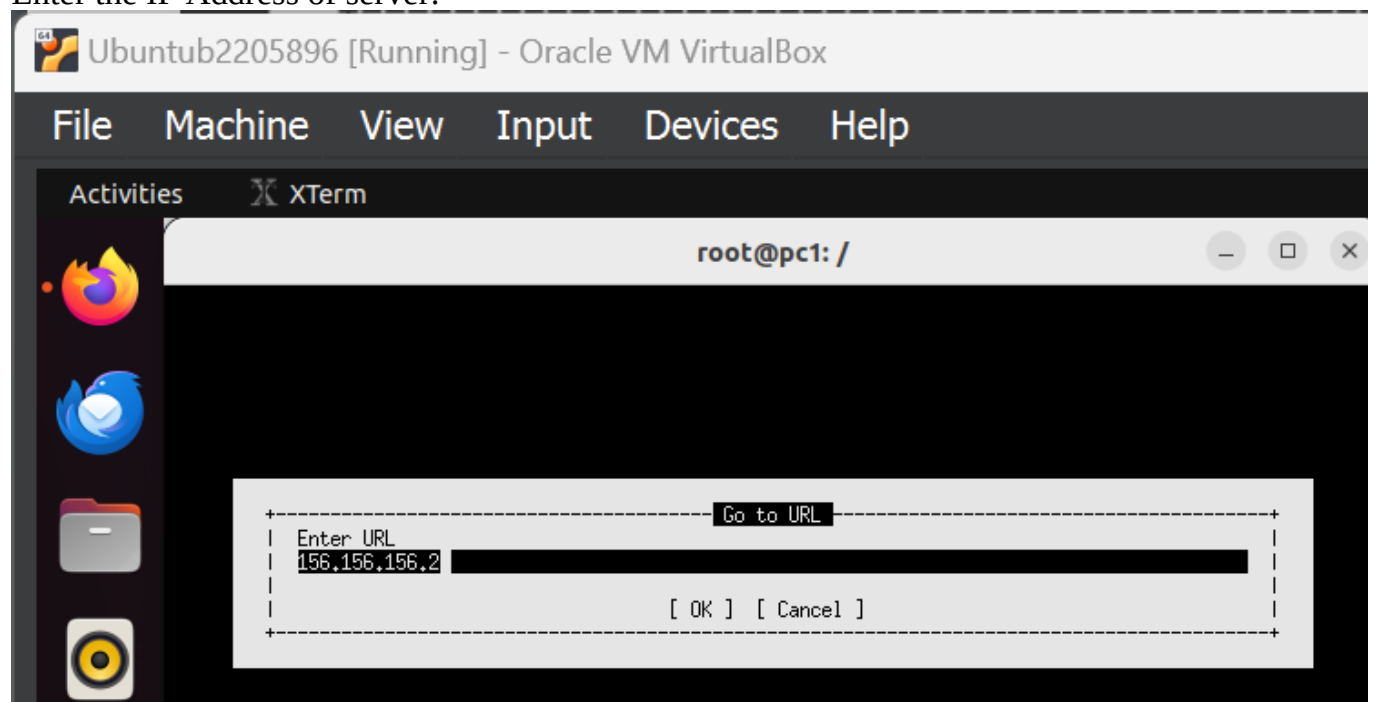
Use **links** command from client:



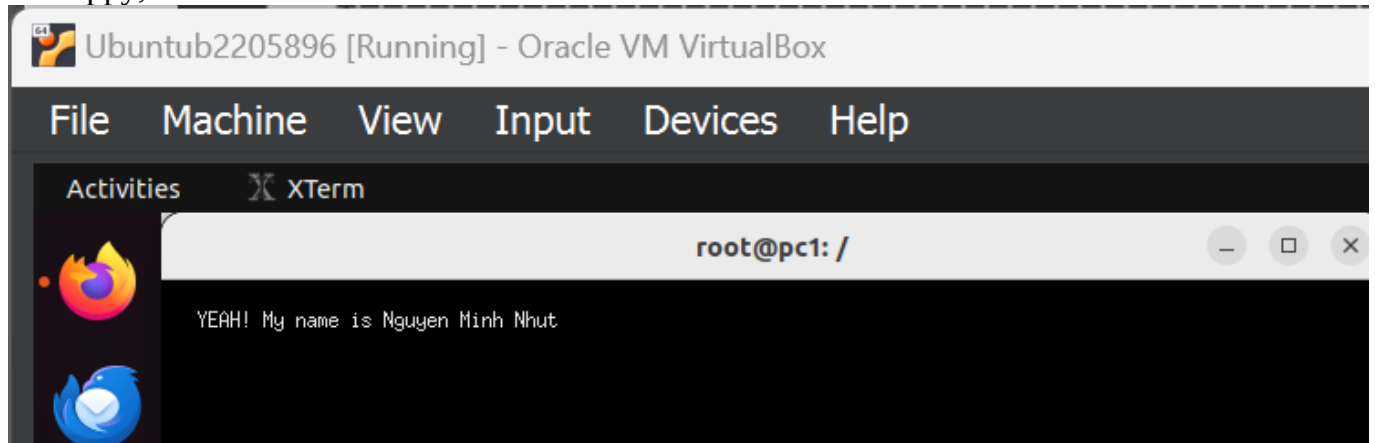
Select “Go to URL”



Enter the IP Address of server:

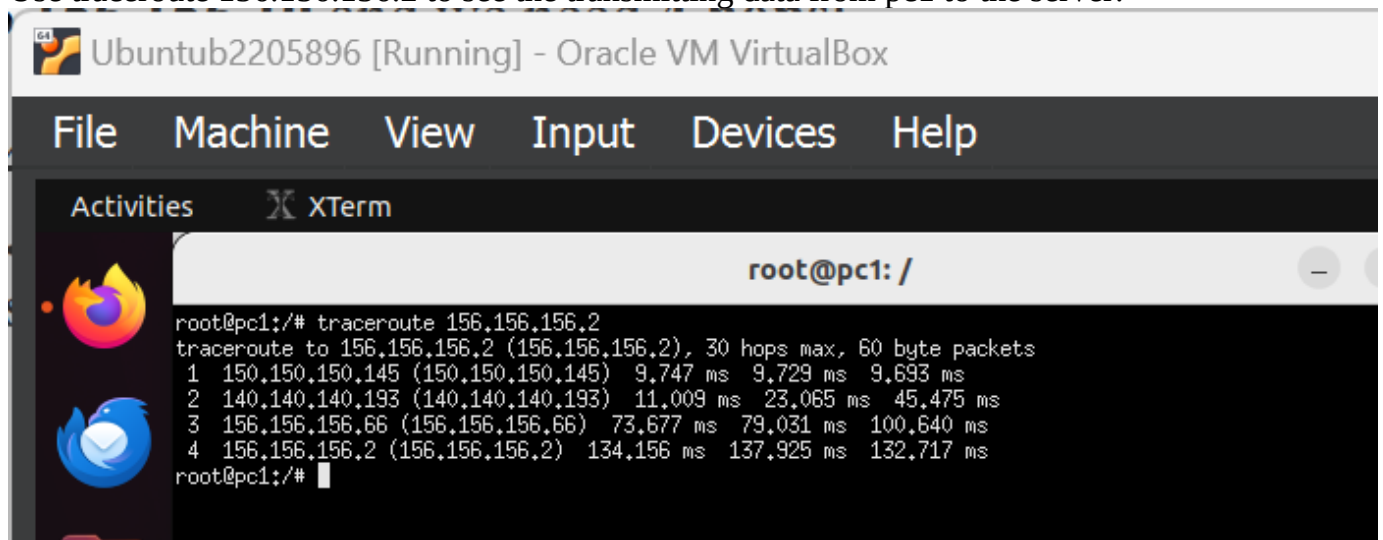


So happy, It worked!!!



**6. The command line to check the hops for transmitting data from PC1 to the web server?
List all hops between PC1 and the Web server.**

Use traceroute 156.156.156.2 to see the transmitting data from pc1 to the server.



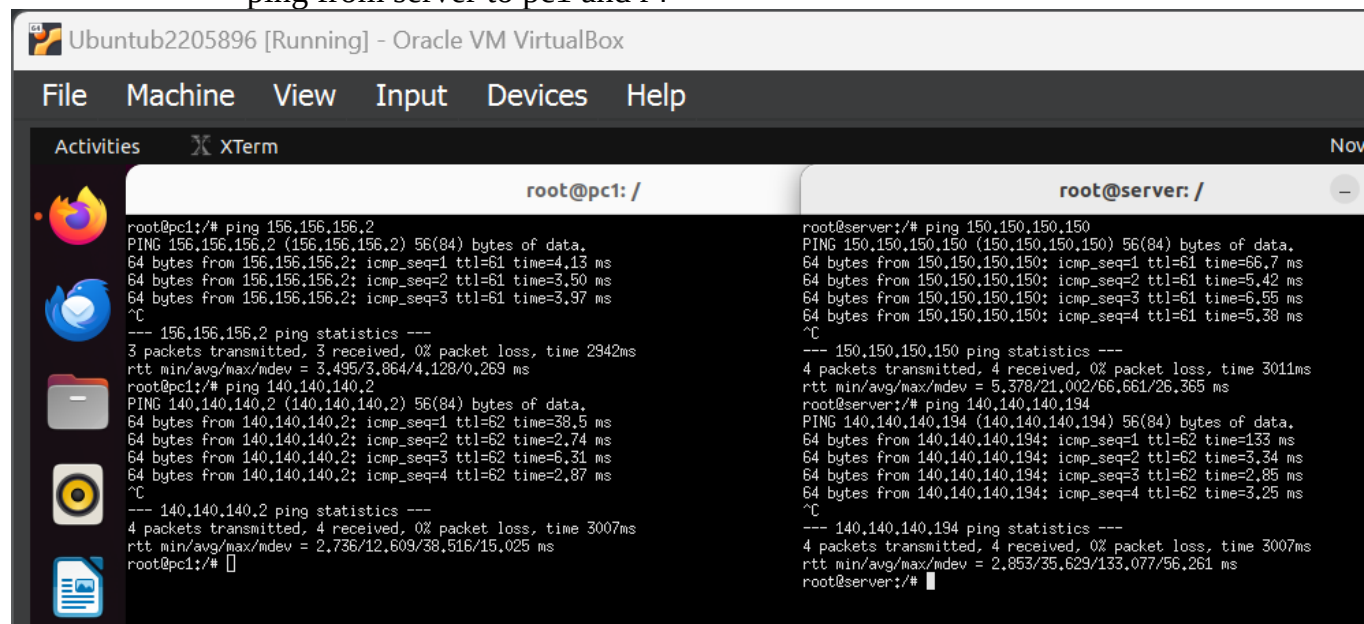
The destination is 156.156.156.2 and we need 4 hops:

- The first next hop is 150.150.150.145 (r4).
- The second is 140.140.140.193 (r1).
- The third is 156.156.156.66 (r2).
- The last is 156.156.156.2 (server).

7. Check the network system constructed (using the ping command)

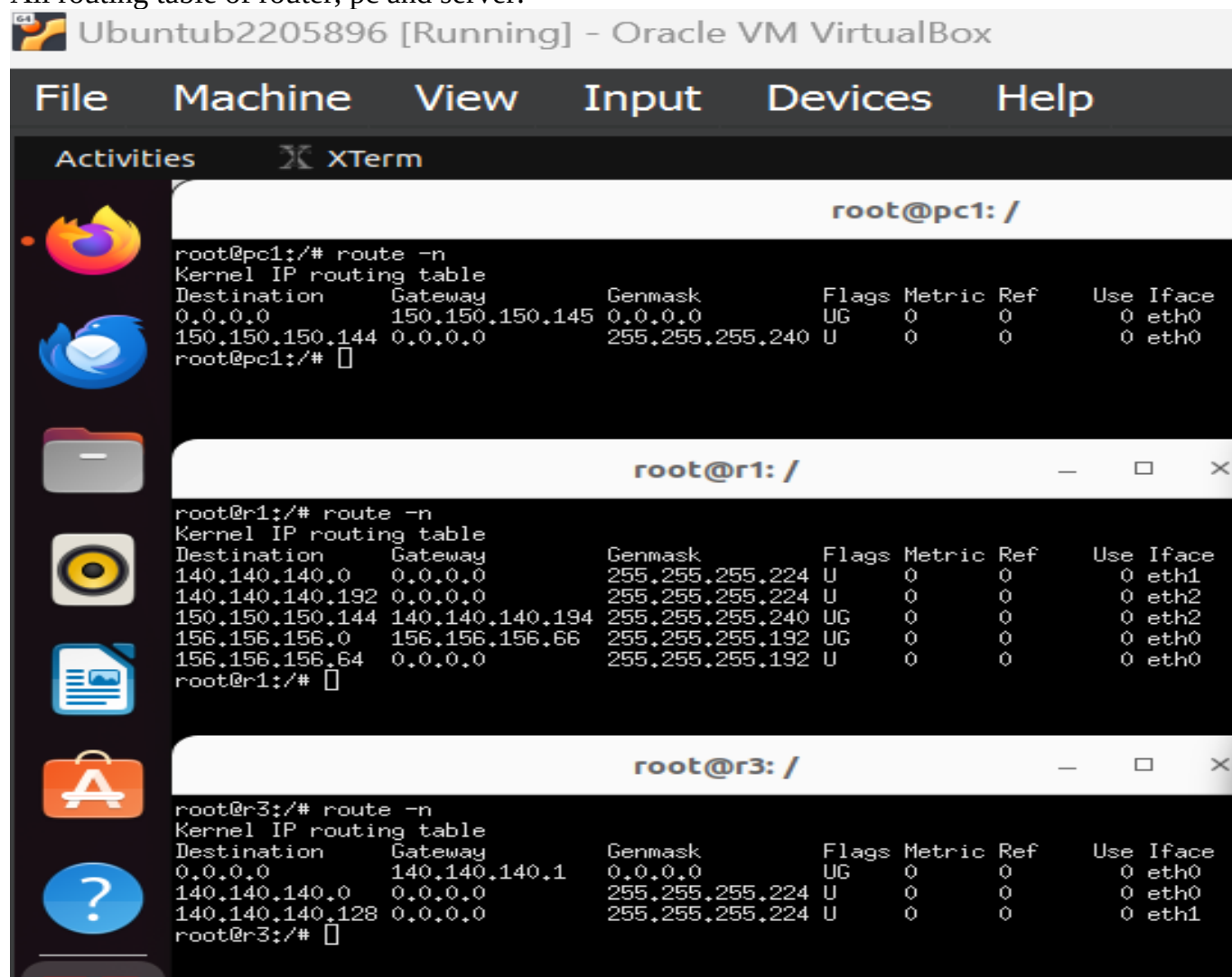
Test connection: ping from pc1 to server and r3

ping from server to pc1 and r4



```
root@pc1: /  
root@pc1:~# ping 156.156.156.2  
PING 156.156.156.2 (156.156.156.2) 56(84) bytes of data.  
64 bytes from 156.156.156.2: icmp_seq=1 ttl=61 time=4.13 ms  
64 bytes from 156.156.156.2: icmp_seq=2 ttl=61 time=3.50 ms  
64 bytes from 156.156.156.2: icmp_seq=3 ttl=61 time=3.97 ms  
^C  
--- 156.156.156.2 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2942ms  
rtt min/avg/max/mdev = 3.495/3.864/4.128/0.269 ms  
root@pc1:~# ping 140.140.140.2  
PING 140.140.140.2 (140.140.140.2) 56(84) bytes of data.  
64 bytes from 140.140.140.2: icmp_seq=1 ttl=62 time=38.5 ms  
64 bytes from 140.140.140.2: icmp_seq=2 ttl=62 time=2.74 ms  
64 bytes from 140.140.140.2: icmp_seq=3 ttl=62 time=6.31 ms  
64 bytes from 140.140.140.2: icmp_seq=4 ttl=62 time=2.87 ms  
^C  
--- 140.140.140.2 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 2.736/12.609/38.516/15.025 ms  
root@pc1:~#  
  
root@server: /  
root@server:~# ping 150.150.150.150  
PING 150.150.150.150 (150.150.150.150) 56(84) bytes of data.  
64 bytes from 150.150.150.150: icmp_seq=1 ttl=61 time=66.7 ms  
64 bytes from 150.150.150.150: icmp_seq=2 ttl=61 time=5.42 ms  
64 bytes from 150.150.150.150: icmp_seq=3 ttl=61 time=6.55 ms  
64 bytes from 150.150.150.150: icmp_seq=4 ttl=61 time=5.38 ms  
^C  
--- 150.150.150.150 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3011ms  
rtt min/avg/max/mdev = 5.378/21.002/66.661/26.365 ms  
root@server:~# ping 140.140.140.194  
PING 140.140.140.194 (140.140.140.194) 56(84) bytes of data.  
64 bytes from 140.140.140.194: icmp_seq=1 ttl=62 time=133 ms  
64 bytes from 140.140.140.194: icmp_seq=2 ttl=62 time=3.34 ms  
64 bytes from 140.140.140.194: icmp_seq=3 ttl=62 time=2.85 ms  
64 bytes from 140.140.140.194: icmp_seq=4 ttl=62 time=3.25 ms  
^C  
--- 140.140.140.194 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 2.853/35.629/133.077/56.261 ms  
root@server:~#
```

All routing table of router, pc and server:



```
root@pc1: /  
root@pc1:~# route -n  
Kernel IP routing table  
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface  
0.0.0.0          150.150.150.145 0.0.0.0         UG        0      0      0 eth0  
150.150.150.144 0.0.0.0         255.255.255.240 U        0      0      0 eth0  
root@pc1:~#  
  
root@r1: /  
root@r1:~# route -n  
Kernel IP routing table  
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface  
140.140.140.0    0.0.0.0         255.255.255.224 U        0      0      0 eth1  
140.140.140.192 0.0.0.0         255.255.255.224 U        0      0      0 eth2  
150.150.150.144 140.140.140.194 255.255.255.240 UG        0      0      0 eth2  
156.156.156.0    156.156.156.66 255.255.255.192 UG        0      0      0 eth0  
156.156.156.64  0.0.0.0         255.255.255.192 U        0      0      0 eth0  
root@r1:~#  
  
root@r3: /  
root@r3:~# route -n  
Kernel IP routing table  
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface  
0.0.0.0          140.140.140.1  0.0.0.0         UG        0      0      0 eth0  
140.140.140.0    0.0.0.0         255.255.255.224 U        0      0      0 eth0  
140.140.140.128 0.0.0.0         255.255.255.224 U        0      0      0 eth1  
root@r3:~#
```

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root@server: /

```
root@server:/# route -n
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface
0.0.0.0          156.156.156.1   0.0.0.0          UG    0      0      0 eth0
156.156.156.0    0.0.0.0          255.255.255.192 U    0      0      0 eth0
root@server:/#
```

root@r2: /

```
root@r2:/# route -n
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface
0.0.0.0          156.156.156.65  0.0.0.0          UG    0      0      0 eth1
156.156.156.0    0.0.0.0          255.255.255.192 U    0      0      0 eth0
156.156.156.64   0.0.0.0          255.255.255.192 U    0      0      0 eth1
root@r2:/#
```

root@r4: /

```
root@r4:/# route -n
Kernel IP routing table
Destination      Gateway          Genmask          Flags Metric Ref    Use Iface
0.0.0.0          140.140.140.193 0.0.0.0          UG    0      0      0 eth0
140.140.140.128  0.0.0.0          255.255.255.224 U    0      0      0 eth1
140.140.140.192  0.0.0.0          255.255.255.224 U    0      0      0 eth0
150.150.150.144  0.0.0.0          255.255.255.240 U    0      0      0 eth2
root@r4:/#
```

--- THE END---